UNIVERSITY OF CAMBRIDGE

PhD Project - 1st Year

Privacy-Preserving and Scalable Occupant Tracking in Indoor Environments

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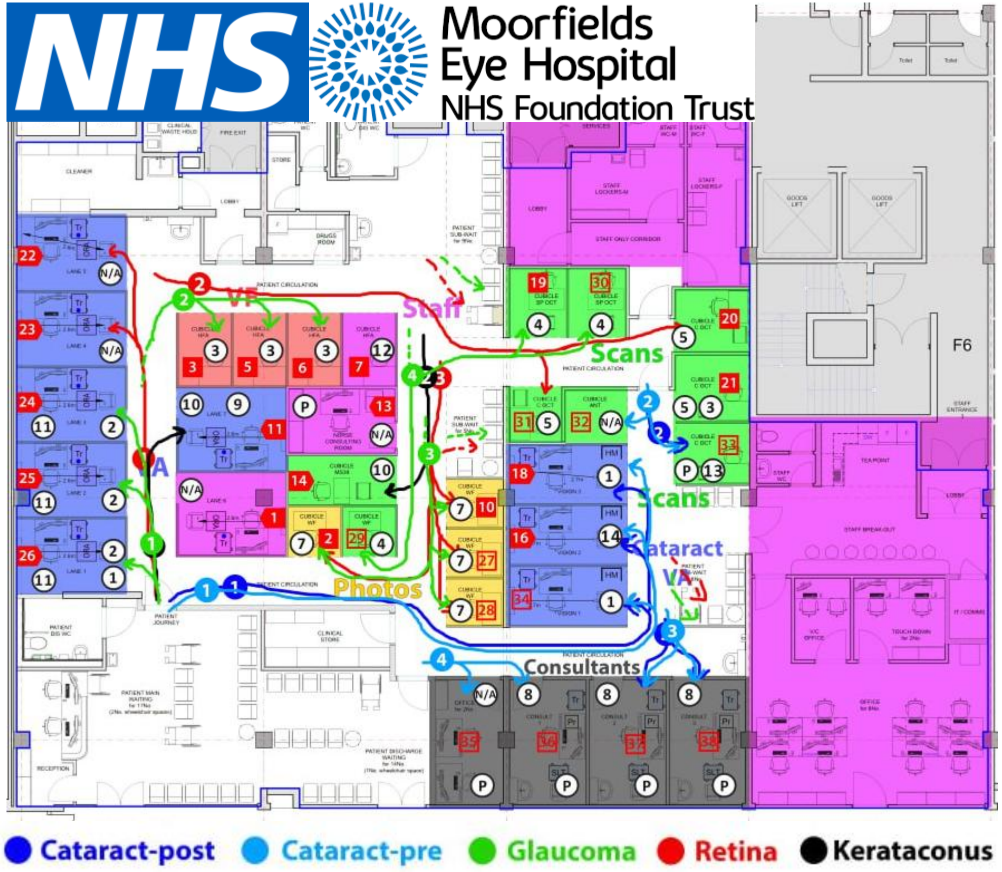
Why occupant tracking?

- **Building optimisation & energy reduction**
- **Shopper behaviour analysis** and queue management.
- **Patient flow optimisation in healthcare facilities**



Patient journeys by cases [3]

Diagnosis cubicles [3]

[3] https://connected-environments.org/projects/hercules/

Ubisense Tracking tags [3]

(Ultra Wideband)

Spatiotemporal data of patient journeys [3]
(involving 4000 patients and 4 clinic layouts iterations)

Result: 9-minute decrease in mean patient visit length [3]

Giving tracking tags to patients…
Guess the **acceptance percentage**?

- Privacy concerns: **Only 50%** of patients accepted carrying a tracking tag
- Not a repeatable process, **hard to deploy** and **scale**

The research question

How to design an occupant tracking system
that delivers spatiotemporal data
while being:

(1) Privacy preserving (2) Scalable
(3) Easy to deploy

Methodology

- **Privacy preservation:**

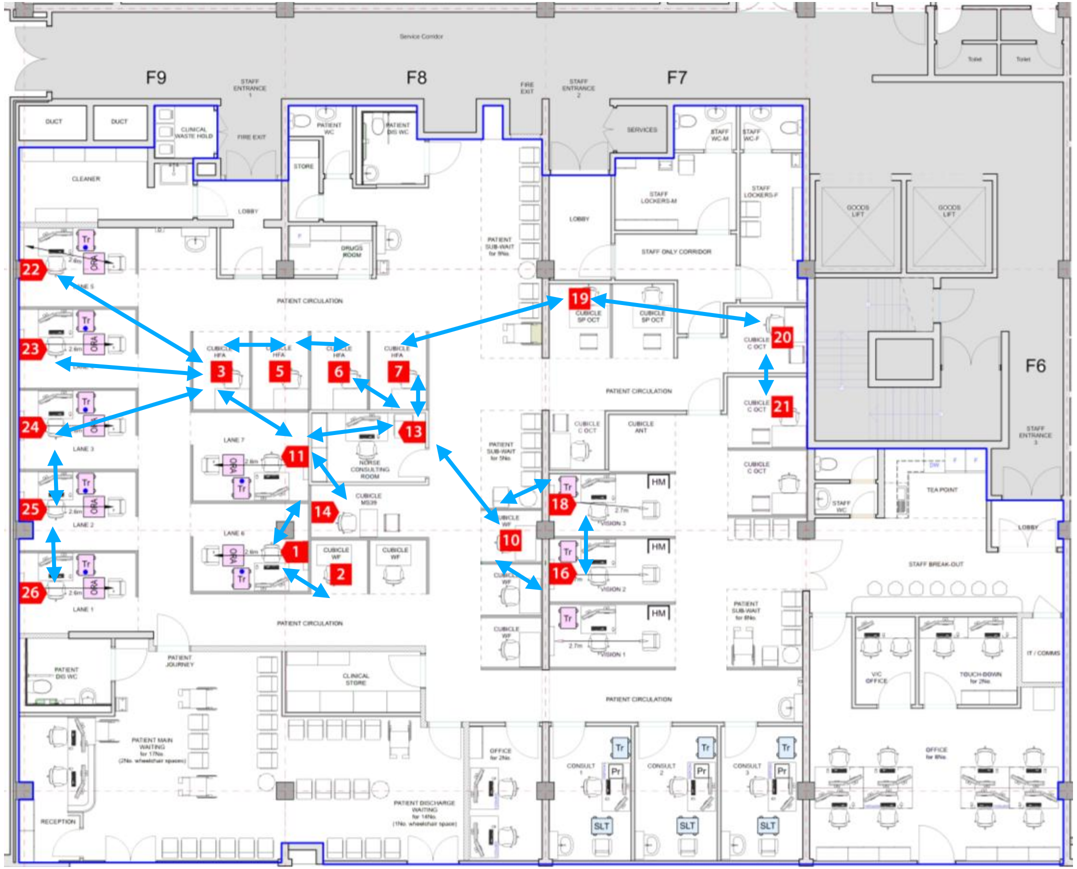
  Edge computing

  → Using mmWave
     Radar-camera fusion

- **Scalability & ease of deployment:**

  Mesh networking

  → Using Private 5G:
     DECT NR+ on 1.9 GHz band



NHS Moorfields London floorplan [3]

Methodology

- **Why Radar-Camera fusion?**

Radar:
- High-privacy areas
- All lighting conditions
- Localisation (depth)

Camera:
- Low-privacy areas
- Better visual features
- Longer range

Best of two worlds!

- **Why Private 5G (DECT NR+ 1.9 GHz)?**

- **No interference**
- High theoretical node density (**1 million/km^2**)
- **Longer range** than 2.4 GHz (Wi-Fi, Thread, Zigbee..)
- **Higher throughput** than 868 MHz (LoRa), **no duty cycle**

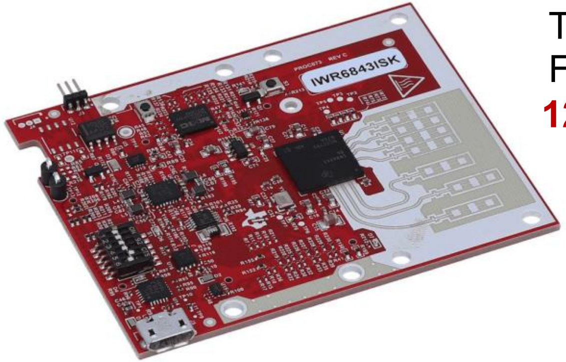
DECT NR+ Sweet spot!

Fisheye cameras

75° FoV

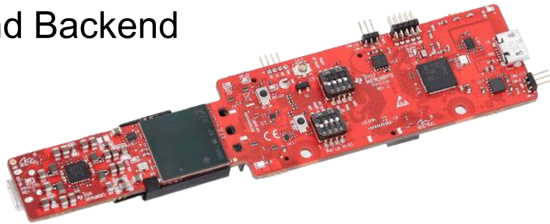
180° FoV - High distortion

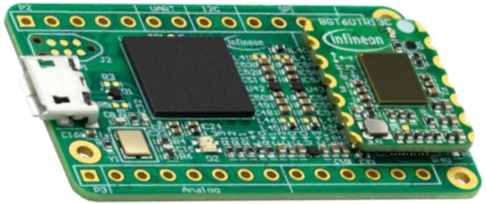
Indoor mmWave radars (60 GHz)



Texas Instruments IWR6843
Frontend and Backend
120° FoV

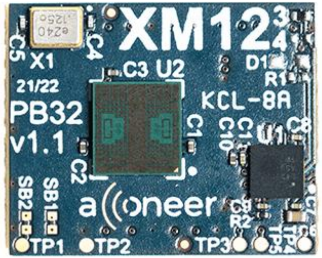
Texas Instruments IWR6843AOPEVM
Frontend and Backend
120° FoV





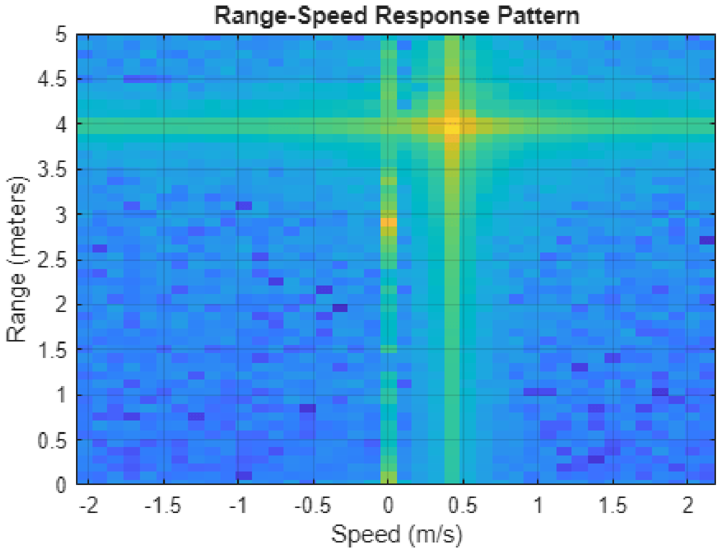
Infineon BGT60TR13C
Only Frontend
120° FoV

Acconeer XM123 – Low power
Frontend and Backend
65° FoV

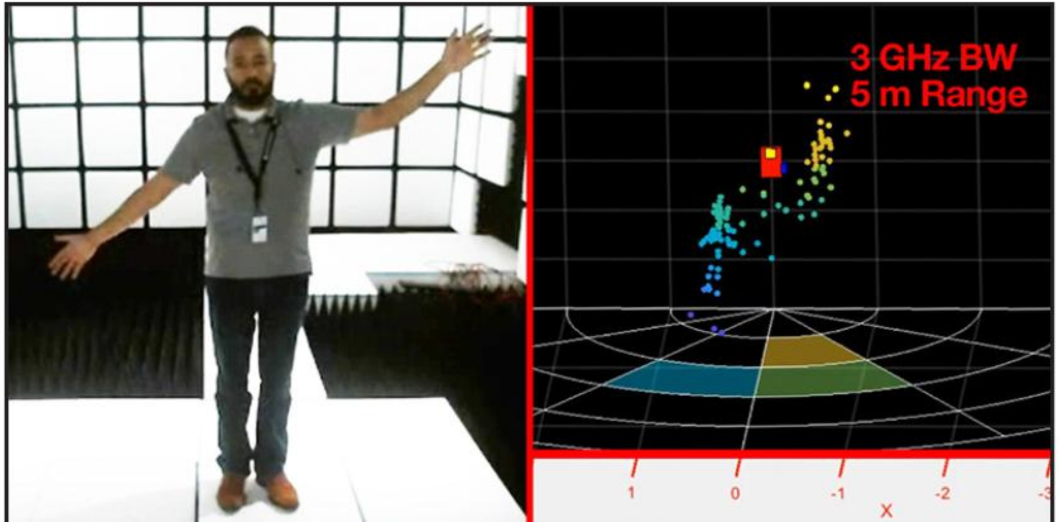


Indoor mmWave radars (60 GHz)

How radar data looks like?

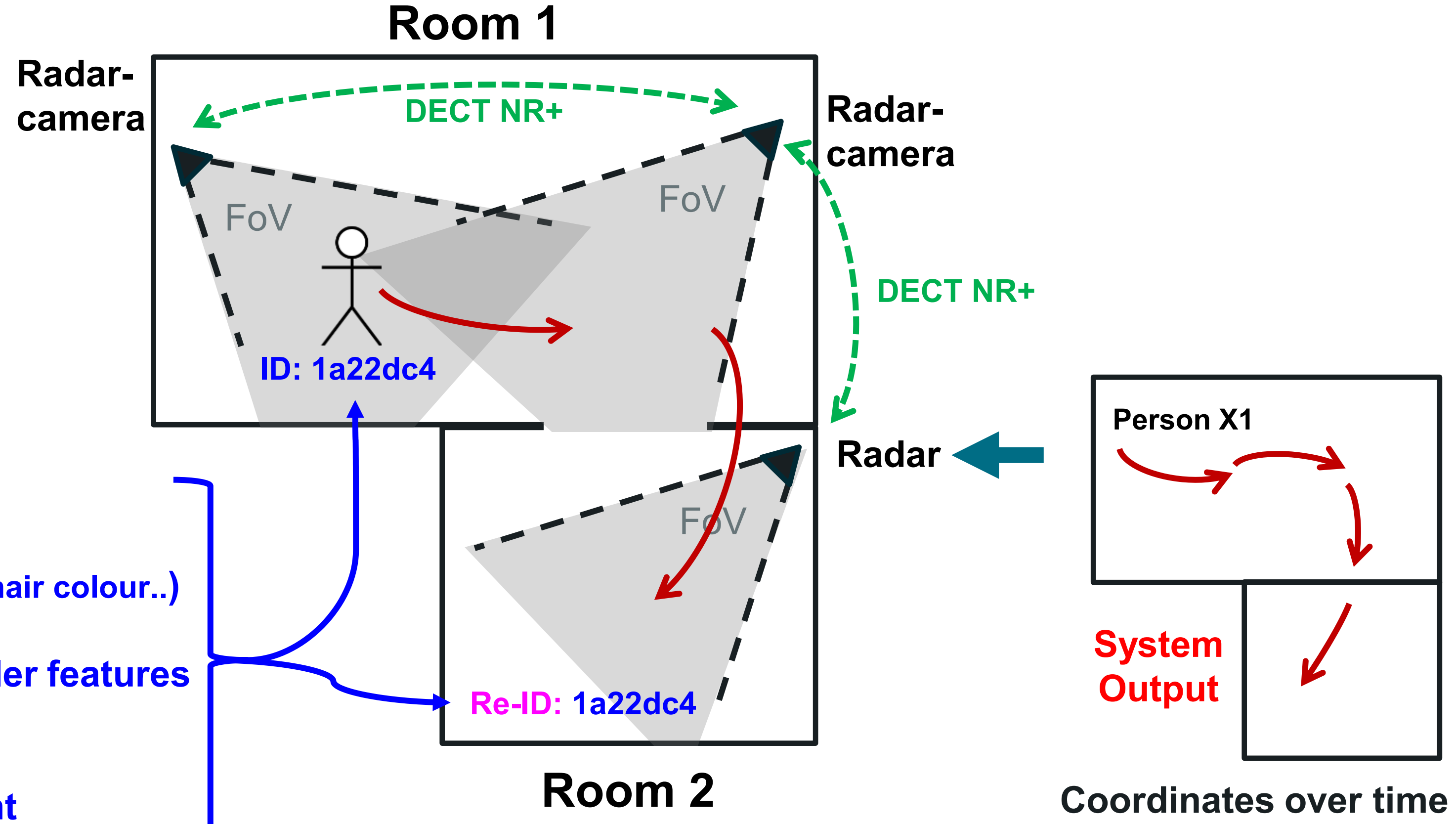


Range-Doppler radar plot [7]



Radar point cloud (post processing) [8]

How it works?



Visual features
- (Gait, clothes colour, hair colour..)

Doppler/Micro-Doppler features
- (Gait, height, speed..)

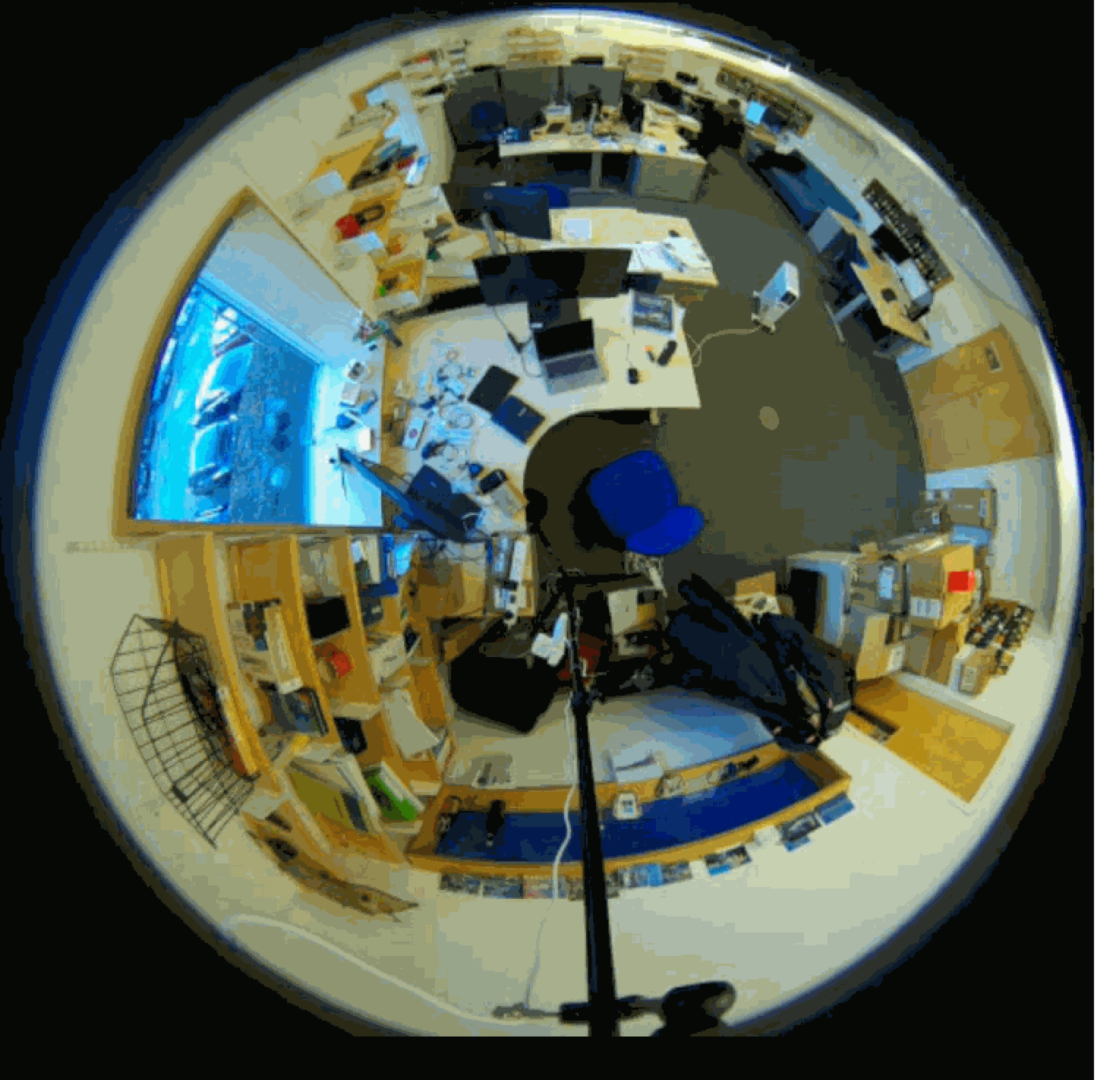
BLE/Wi-Fi Fingerprint

Fisheye cameras: Pipeline

180° image stream → Unrolling to un-distort → Run normal CNN (i.e. YOLO) → Detected objects

180° image stream → Run modified CNN with OBB* → Detected objects

*OBB: Oriented Bounding Boxes



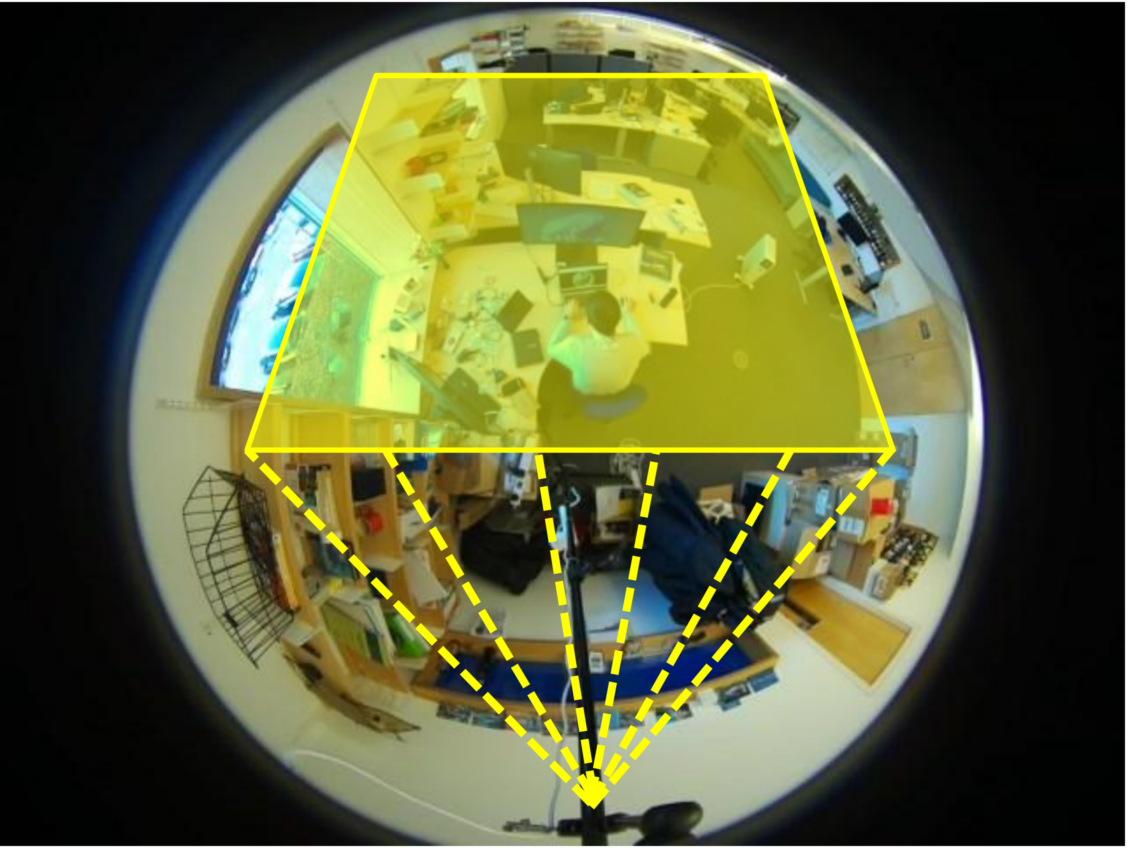
Fisheye cameras

OBB (Oriented Bounding Boxes) in action

YOLO26-OBB is selected [6]

Retraining and optimisation in progress

Radar-Camera Fusion Challenges



- Mapping radar depth to 2D optical images.
- Aligning spatial coordinates across overlapping FoVs.

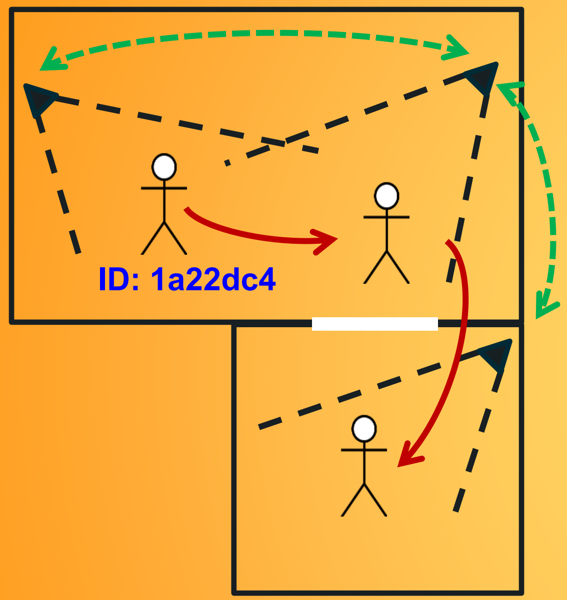
DECT NR+ Mesh challenges

Based on Nordic nRF9151

- Using relatively new DECT NR+
- Implementing custom mesh on the PHY layer.
- Working with OFDM (Orthogonal Frequency-Division Multiplexing) and OTA sync.

References:

[1] https://github.com/RizwanMunawar/yolov7-object-tracking?tab=readme-ov-file
[2] https://www.retailsensing.com/people-counting/track-the-customer-journey-and-maximise-retail-store-sales/
[3] https://connected-environments.org/projects/hercules/
[4] https://github.com/ucl-casa-ce/hercules-web
[5] https://github.com/duducosmos/defisheye
[6] https://docs.ultralytics.com/models/yolo26/
[7] https://www.mathworks.com/help/radar/ug/iq-data-collection-detection-application-example.html
[8] https://www.ti.com/lit/pdf/spry328



Thank you

Don't forget to check the poster :)

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